

CSE / EEE / ETE 499A (Section 1)

Ethical and Professional Responsibility Report (CO6)

Project Title:

ShopMate — AI A Catalog-Grounded, Omnichannel AI Sales & Support Agent for Facebook, Instagram, WhatsApp, and Website Commerce

Submitted To

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Ethical and Professional Responsibility in Project Development

1. Introduction

ShopMate AI is a multi-tenant, catalog-grounded, omnichannel AI sales-and-support agent that unifies Facebook Messenger, Instagram, WhatsApp Business, and a website widget into a single merchant inbox, generating customer replies via Retrieval-Augmented Generation (RAG) grounded in the merchant's own product catalog, detecting purchase intent and complaints, and creating orders directly from chat.

Ethics and professionalism matter to this project specifically because it inserts an autonomous AI system directly into a financial transaction path between a merchant and a customer: an AI-authored message can state a price, promise stock, or commit to a delivery timeline on the merchant's behalf, and it processes real customer personal data (names, phone numbers, order details) across four external platforms the team does not control. A design or process failure here is not merely a bug — it can cause direct financial harm to a customer, reputational harm to a merchant, or a data-privacy breach, which is why ethical and professional responsibility is treated as a first-class engineering requirement rather than an afterthought.

This report covers the full seven-month development lifecycle spanning CSE499A and CSE499B — problem definition and requirement analysis, system and database design, implementation, testing and validation, and deployment/demo — and should be read alongside the CO1 proposal, CO2 design report, and CO4 impact and constraints report, which this report cross-references rather than duplicates.

2. Relevant Ethical Principles

- **Honesty and non-deception (truthfulness of AI output)** — the system must never present a fabricated price, stock level, or delivery promise as fact; this is the project's central ethical design constraint and is enforced technically, not just policy-wise, by the Grounding Guardrail described in the CO2 design report (Section 3.2–3.3).
- **Privacy** — customer names, phone numbers, and order data are personal information that must be collected, stored, and processed with informed consent and minimal exposure, consistent with the data-handling commitments already made in the CO1 proposal (Section 3.6) and CO4 report (Section 3.3).
- **Safety and avoidance of harm** — an AI-authored commitment a merchant cannot honor is a real (if modest) financial harm to a customer; the system is designed so a human remains in the loop before any AI draft reaches a customer by default (CO2, Section 2).
- **Fairness and inclusivity** — the system must serve Bangla-, English-, and Banglish-speaking customers with comparable accuracy rather than privileging English speakers, since language-based accuracy disparities would functionally exclude a large share of the target market from good service.
- **Accountability** — every AI-generated action (a reply, an order, an escalation) must be traceable to a specific model version, prompt, and retrieved context, so that when something goes wrong, responsibility can be assigned and the failure understood rather than treated as an unexplainable “black box” error.
- **Professional competence** — the team commits to only claiming capabilities the system actually has (e.g., not advertising accuracy or language coverage the AI subsystem has not been evaluated against), consistent with the IEEE and ACM codes of ethics referenced below.

Professional Standards Referenced

Two established professional codes of ethics ground this analysis:

- **IEEE Code of Ethics** — commits engineers to holding paramount the safety, health, and welfare of the public, to being honest and realistic in stating claims or estimates based on available data, and to treating fairly all colleagues and the public regardless of factors such as race, religion, or national origin [1] — directly relevant to the grounding-rule and Bangla/Banglish fairness principles above.
- **ACM Code of Ethics and Professional Conduct** — its first general principle requires computing professionals to contribute to society and human well-being while minimizing negative consequences, including threats to health, safety, and privacy, and states that when the interests of multiple groups conflict, the needs of those less advantaged should be given increased attention [2] — directly relevant to prioritizing the underserved Bangla/Banglish-speaking customer segment (Section 3 below) and to the privacy commitments in Section 4.2.

3. Stakeholder Analysis

Stakeholder	Expectations / Concerns	Potential Ethical Conflict
Customers	Accurate, fast, honest answers; their personal data (name, phone, order history) handled responsibly; not misled by an AI they may not realize is automated.	Speed/automation vs. accuracy — a faster AI-drafted reply is only ethical if it is also correct; over-trusting an AI reply could cause the customer financial harm if the grounding rule fails.
Merchants (primary client persona)	The AI represents their brand faithfully, never overpromises on their behalf, and does not leak their catalog/pricing data to competitors (a multi-tenant risk).	Automation vs. control — merchants want efficiency gains from automation but also want final say over customer-facing commitments, which is why AI drafts require human approval by default (CO2, Section 2).
Development team	Build a system they can defend as safe and honest under academic and, potentially, future commercial scrutiny.	Feature ambition vs. safety — pressure to ship more automated capability (e.g., full auto-send) faster can conflict with the more conservative human-in-the-loop default.
Society / underserved language communities	Bangla/Banglish-speaking customers receive service quality comparable to English speakers, rather than being treated as a lower priority.	Resource allocation — English-first NLP tooling is more mature and cheaper to build with; prioritizing Bangla/Banglish quality (per the ACM Code's “less advantaged” principle) can mean slower development or higher cost.
Regulators (data protection, consumer protection)	Compliance with applicable data-protection and consumer-protection law (CO4, Section 4).	Innovation speed vs. compliance overhead — a student capstone team has limited legal resources to fully verify compliance before any real commercial launch.
Platform providers (Meta/WhatsApp, LLM provider)	Compliant use of their APIs and policies (messaging windows, business verification, acceptable-use terms).	Product ambition vs. platform policy — features that would improve the product (e.g., more aggressive proactive outreach) may conflict with platform messaging-window rules.

4. Ethical Considerations Across Development Phases

4.1 Problem Definition & Requirement Analysis

The problem the project addresses — automating merchant customer service — carries an inherent ethical implication: automation that fails silently or inaccurately does more harm than the manual process it replaces,

since customers may trust an instant, confident-sounding AI reply more than they would a visibly overworked human agent. This shaped the requirement-analysis decision (CO1 proposal, Section 3.5) to make catalog-grounded accuracy a must-have success metric rather than a nice-to-have, and to define Bangla/Banglish support as a first-class requirement rather than an English-first product with translation bolted on, directly addressing the fairness/inclusivity principle in Section 2.

4.2 Design Phase

- **Safety** — the Grounding Guardrail (CO2, Section 3.2) is a design-level safety mechanism, not a testing-phase check: it is architecturally positioned between the LLM Gateway and the Delivery Worker so an ungrounded claim structurally cannot reach a customer.
- **Privacy-by-design** — tenant_id-scoped row-level security and pre-filtered vector search (CO2, Section 3.3, 4.3) were chosen at the schema-design stage specifically so that cross-tenant data leakage is structurally impossible rather than merely policy-forbidden — a direct application of privacy-by-design.
- **Secure-by-design** — webhook signature verification and idempotency at the Ingestion Gateway (CO2, Section 3.2) were designed as the system's single trust boundary from the outset, rather than added after an incident.
- **Environmental/societal impact** — the tiered model-routing design (CO2, Section 6.6) was chosen partly for cost control and partly to reduce the energy footprint of routine AI calls, as discussed in the CO4 report (Section 2.1).

4.3 Implementation Phase

- **Responsible use of tools, data, and resources** — only seeded, synthetic merchant and customer data is used during development and internal testing; no real customer PII is processed before appropriate consent and data-handling processes are in place (CO1, Section 3.6; CO4, Section 3.3).
- **Intellectual property** — all third-party libraries are reviewed for open-source license compatibility (CO4, Section 4, legal table) before inclusion; all code, diagrams, and written content in the CO1–CO6 reports are the team's own original work, and every external claim, statistic, or technique referenced in these reports is cited to its source (see CO1, Section 2 literature review and References) rather than presented as the team's own finding.

4.4 Testing & Validation

- **Ethical testing practices** — all functional and load testing (CO5, Section 9) uses seeded synthetic conversations and orders; no real customer is contacted or experimented on without their knowledge, and no production merchant traffic is used as an unwitting test population.
- **Data integrity and transparency in reporting** — the success metrics defined in the CO1 proposal (Section 3.5) and computed in the CO2 report (Section 6.7) — automation rate, order uplift, response time — are reported as measured against the seeded test catalog, and this report and its companions are explicit that no field/production data has yet validated these figures at commercial scale, avoiding the ethical failure of presenting prototype-stage estimates as proven real-world results.

4.5 Deployment & Maintenance

- **User safety and long-term impact** — the human-in-the-loop default (AI drafts require an explicit “Send Now”, CO2 Section 2) is deliberately kept as the default even at deployment, rather than being a training-wheels feature removed once the model “seems reliable,” since reliability at demo scale does not guarantee reliability at production scale.

- **Accountability and monitoring** — every AI-generated message is flagged `ai_generated` in the schema (CO2, Section 4.2) and every AI usage event is logged (`usage_events` table), which supports both cost accountability (CO5, Section 11) and incident investigation if an AI response causes a complaint.
- **Plan for handling failures or misuse** — a failed grounding check triggers a bounded retry and then a safe fallback template rather than a silent failure (CO2, Section 3.2); complaints are always routed to a human rather than auto-resolved (CO1, Section 3.5 must-haves), giving merchants a clear escalation path when the AI gets something wrong.

5. Legal and Regulatory Compliance

This section summarizes the compliance items most relevant to ethical/professional conduct; the full standards and regulations table is provided in the CO4 Project Impact and Constraints Report (Section 4) and is not duplicated here in full.

- **Data protection** — Bangladesh's data protection framework (as currently in force/draft) and GDPR-aligned best practice are both used as a baseline (CO4, Section 4); the multi-tenant row-level security design (Section 4.2 above) is the primary technical control.
- **Consumer protection** — the Bangladesh Consumer Rights Protection Act, 2009 provisions on misleading claims motivate the hard grounding rule (CO4, Section 4) — an AI-quoted price or promise is, legally as well as ethically, a representation the merchant may be held to.
- **Platform policy compliance** — Meta Platform Terms and the WhatsApp Business Messaging Policy, including the 24-hour customer-service messaging window, are enforced at the Delivery Worker (CO2, Section 3.3; CO4, Section 4).
- **Payment/financial** — no raw card data is stored in-house; checkout is delegated to a PCI-DSS-compliant third-party payment gateway (CO4, Section 4).

6. Risk Assessment and Mitigation

Ethical Risk	Likelihood	Severity	Mitigation / Contingency
AI states an incorrect price/stock/delivery promise (hallucination)	Medium	High	Grounding Guardrail rejects ungrounded claims (CO2, 3.2); bounded retry then safe fallback template; human approval required before send by default.
Cross-tenant data leakage	Low	High	Row-level security + tenant-scoped vector search (CO2, 3.3/4.3); code review checklist includes a tenant-isolation check before merging any new query path.
Bangla/Banglish customers receive systematically worse service than English speakers	Medium	Medium	Dedicated language-ID step and Bangla-specific evaluation set (CO1 literature review, Section 2.2); accuracy tracked per-language, not only in aggregate.
Over-reliance on AI reduces merchant human oversight over time (“automation complacency”)	Medium	Medium	Human-in-the-loop remains the default even post-launch (Section 4.5); complaints are never auto-resolved.

Use of real customer data before consent/compliance processes are finalized	Low	High	Seeded synthetic data only during the capstone; any future real-data pilot requires an explicit data-processing and consent review before launch.
Misrepresenting prototype-stage metrics as validated real-world results	Low	Medium	All reported metrics are explicitly labelled as measured against the seeded test catalog (Section 4.4); no production-scale claims are made without production-scale data.

7. Professional Responsibility

- **Team roles and accountability** — each of the four functional roles defined in the CO1 proposal (Section 4.5) and CO5 report (Section 3.2) — Backend/Infrastructure, AI/ML, Commerce & Channel, and Frontend/QA — carries direct accountability for the ethical properties of their component (e.g., the AI/ML lead is accountable for grounding-guardrail correctness); the rotating Project Manager role ensures no single member is both building and unilaterally reporting progress.
- **Communication with stakeholders** — progress and limitations are reported honestly to the supervisor at biweekly meetings (CO5, Section 10) rather than only reporting successes; known limitations (e.g., features still on the nice-to-have backlog, CO1 Section 3.5) are disclosed rather than implied to be complete.
- **Honesty and integrity in reporting results** — all performance figures in this report and its companions are described as targets or prototype-stage measurements against a seeded test catalog, not as production-validated claims (Section 4.4), and all external facts, statistics, and techniques are cited to their source rather than presented as original findings.
- **Time management and commitment to quality** — the sprint-based schedule and Gantt chart (CO5, Section 6) intentionally build in an integration-testing and evaluation period before both the CSE499A and CSE499B milestones, rather than treating testing as whatever time happens to remain after development.

8. Reflection and Lessons Learned

The most significant ethical challenge the team anticipated in designing this system was the tension between automation ambition and trustworthiness: an AI sales agent is only valuable to a merchant if it is fast and largely autonomous, but full autonomy is precisely what makes an ungrounded or biased response dangerous. The decision to make human-in-the-loop approval the default (rather than an opt-in safety feature) and to make grounding a hard architectural constraint rather than a prompt-level suggestion (CO2, Section 3.2–3.3) was made specifically to resolve this tension in favor of trustworthiness over pure automation speed, even though it means the system is, by design, less “autonomous” than some competitor tools that auto-send replies without merchant review.

A second lesson concerns scope discipline: the temptation to prioritize the larger, better-resourced English-language NLP ecosystem over the harder, more locally-relevant Bangla/Banglish problem was recognized early and deliberately resisted, following the ACM Code's principle of prioritizing the needs of less-advantaged or underserved groups [2]. In a future project, the team would build a labelled Bangla/Banglish evaluation set even earlier in the schedule, since language-fairness evaluation was identified as important later than it ideally should have been relative to the rest of the design process.

9. Conclusion

Across problem definition, design, implementation, testing, and deployment planning, ShopMate AI's development has treated honesty of AI output, data privacy, fairness across languages, and human accountability as hard engineering constraints rather than aspirational values — most concretely through the architectural Grounding Guardrail, tenant-scoped data isolation, and a human-approval-by-default inbox workflow. The team's professional responsibility has been exercised through accountable role ownership, honest reporting of prototype-stage (rather than production-validated) results, and disciplined adherence to the in-scope/out-of-scope boundaries defined in the CO5 report, consistent with the IEEE and ACM codes of ethics referenced in Section 2.

10. References

- [1] IEEE, "IEEE Code of Ethics," IEEE Policies, Section 7.8. [Online]. Available: <https://www.ieee.org/about/corporate/governance/p7-8>
- [2] ACM, "ACM Code of Ethics and Professional Conduct," Association for Computing Machinery, 2018. [Online]. Available: <https://www.acm.org/code-of-ethics>
- [3] Government of the People's Republic of Bangladesh, Consumer Rights Protection Act, 2009.
- [4] Government of the People's Republic of Bangladesh, Bangladesh Data Protection Act (as in force / draft per latest government notice).
- [5] Meta Platforms, Inc., "WhatsApp Business Messaging Policy," and "Meta Platform Terms." [Online]. Available: <https://business.whatsapp.com/policy>
- [6] PCI Security Standards Council, "Payment Card Industry Data Security Standard (PCI-DSS)." [Online]. Available: <https://www.pcisecuritystandards.org>

Additional supporting literature (COI References 3–8 on RAG-grounding and Bangla/Banglish NLP) is provided in full in the COI Project Proposal and is not duplicated here.